

POLARIS DC-1 — ELECTRICAL: HV, LV, UPS AND GENERATORS

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, ratings and figures are fictional.

1. Scope

- 1.1 This specification covers the design, supply, installation, testing and commissioning of the electrical power system for Polaris DC-1, from the 110 kV incoming termination through the 110/20 kV primary substation, the 20 kV site ring, hall substations, standby generation, UPS and battery systems, to the 400 V busway distribution serving the white space.
- 1.2 The Subcontractor shall carry design responsibility for the means of compliance with the performance requirements of PDC-ER-001 §3, using the sizing basis set out in PDC-BOD-001 §5 as the design intent to be carried forward unchanged unless a design change is agreed under the S1–S3 review process.
- 1.3 Excluded: the 110 kV overhead line and the network operator’s own terminations (Employer’s battery limit, PDC-ER-001 §1.3); IT racks, servers and rack PDUs beyond the busway tap-off (Tenant-supplied, PDC-TA-001 §11); BMS/DCIM field devices and software (refer PDC-SPEC-07).
- 1.4 The electrical system shall be delivered in four Sections aligned with PDC-ER-001 §7, with all shared infrastructure (substation, both transformers, all eight generators) complete and commissioned to full 10 MW capacity as part of Section 1.

2. Standards

Reference	Application
National Building Code of Finland	Electrical services chapters
Land Use and Building Act (132/1999)	Statutory design basis
SFS 6000 series	Low-voltage installations (400 V distribution, busway, final circuits)
SFS 6001	High-voltage installations (110 kV, 20 kV switchgear and substation)
Rescue Act (379/2011)	Fire safety, emergency power for life-safety systems
EN 50600 series	Data centre facility and power distribution design reference
EN 50600-4-2	PUE calculation basis
IEC 61439	Low-voltage switchgear and controlgear assemblies
IEC 62271	High-voltage switchgear and controlgear
IEC 60909	Short-circuit current calculation
IEC 62053-22	Static electricity meters, Class 0.5S — per-rack metering accuracy
IEEE 519 / EN 50160	Power quality — harmonic distortion and voltage characteristics

EN 62040 series	UPS system requirements
Uptime Institute Tier III	Design benchmark for concurrent maintainability (not certified)

Where standards conflict, the more onerous requirement governs, per PDC-BOD-001 §1.2.

3. Equipment schedules

3.1 High-voltage and transformers

Item	Requirement
Incoming supply	110 kV, terminating within the primary substation building (Contractor's battery limit)
Transformers	2 × 110/20 kV, 16 MVA each, ONAN/ONAF, N+1 (either transformer alone carries full site peak)
Site ring	20 kV, running the full length of the shared MEP spine, feeding a dedicated substation at each of the four halls
Switchgear	SF6-free or SF6 gas-insulated switchgear at 110 kV and 20 kV per Engineer approval, metal-clad, arc-fault contained
Earthing	Common bonding network, TN-S, single site earth reference bar in the primary substation building

3.2 Standby generation

Item	Requirement
Generators	8 × 2.5 MVA diesel, 20 MVA installed
Redundancy	N+1 for full site load including cooling — 7 duty (17.5 MVA firm) against a site peak of approximately 16.06 MVA, 1 standby
Location	Enclosed compound, east end of site, away from the north-east residential noise receptor
Fuel storage	48 hours at full site load, bunded, with a fuel polishing system
Load acceptance	Full block load pick-up within 10 seconds of loss of incoming supply
Delivery Section	Complete and commissioned, sized for the full 10 MW site, as part of Section 1
Exhaust and acoustic	Attenuated to meet the 45 dB(A) night-time limit at the north-east boundary

3.3 UPS and batteries

Item	Requirement
Quantity	16 modules total — 4 per hall, 1,000 kVA each
Configuration per hall	N+1 (3 duty + 1 standby), giving 3.0 MVA firm capacity against a required approximately 2.63 MVA at 0.95 output power factor
Autonomy	5 minutes at full load
Type	Static, double-conversion
Battery chemistry	Lithium-ion (LFP), in a dedicated fire-rated battery room per hall
Segregation	Physical segregation between UPS modules so a single-module event cannot propagate
Fire detection	Off-gas / thermal-runaway detection ahead of the aspirating smoke system, refer PDC-SPEC-07

3.4 Distribution

Item	Requirement
Hall substation	20 kV to 400 V step-down, one per hall

White space distribution	Overhead busway, no raised floor, coordinated within the 2.4 m service zone
Redundancy	N+1 at every level between incoming transformer and rack PDU
Metering	Per-hall and per-feed energy metering, per-rack metering Class 0.5S (IEC 62053-22), reporting to the DCIM

4. Installation and workmanship

4.1 All 110 kV and 20 kV switchgear shall be installed by SFS 6001-competent personnel and tested in accordance with the manufacturer's factory and site test procedures before energisation.

4.2 Busway shall be installed within the coordinated overhead service zone, with tap-off boxes positioned to suit the rack layout without penetrating fire compartment walls except through fire-stopped, EI 120-rated sleeves.

4.3 Cable containment between the primary substation, the generator compound and each hall substation shall run within the shared MEP spine, with sub-compartmentation at each hall's connection point per PDC-BOD-001 §3.4, so that a single hall's plant room cannot compromise the spine serving the other three halls.

4.4 Generator exhaust routing and acoustic enclosures shall be designed and installed to demonstrate compliance with the 45 dB(A) night-time limit at the north-east boundary, with acoustic modelling submitted at S2.

4.5 Battery rooms shall be installed with chemical-resistant flooring, dedicated ventilation sized for thermal-runaway off-gas, and segregation between modules such that fire cannot propagate module-to-module within the room.

4.6 Each hall's electrical infrastructure shall be installed and commissioned so that it is complete and independently testable at that hall's Section handover, without requiring access to an operating hall's plant.

4.7 All LV switchgear and distribution boards shall comply with SFS 6000 and IEC 61439, with form of separation confirmed at S2 detailed design.

5. Control and sequences of operation

5.1 Loss-of-utility sequence

Step	Event	Time
1	Loss of 110 kV incoming supply detected	T+0
2	UPS assumes load, no perceptible transient at rack PDU	T+0 to T+10s
3	Generator start signal issued, no single point of failure in the start control system	T+0
4	Generators reach rated speed and voltage, automatic transfer switch closes	Within 10 seconds of loss of supply
5	UPS reverts to normal float/charge mode once generator supply is stable	After transfer complete

5.2 Power quality requirements

Parameter	Requirement
Voltage THD at rack PDU	≤ 3% (IEEE 519 / EN 50160 basis)
Current THD at rack PDU	≤ 5% under Tenant's typical IT load profile
Voltage tolerance	± 5% of nominal sustained; ± 10% for transient events not exceeding 100 ms
Frequency tolerance	± 1% of nominal (50 Hz)
Transfer time, utility loss to UPS-carried load	No perceptible transient at the rack PDU
Transfer time, UPS to generator-carried load	Within 10 seconds, UPS bridging the interval

5.3 Protection and discrimination

5.3.1 The Subcontractor shall carry out a full protection coordination and discrimination study across all levels of the electrical system — from the 110 kV incoming protection, through the 20 kV ring, hall substation transformers, UPS input/output, busway and final distribution to the rack PDU — such that a fault at any level clears by the protective device immediately upstream of the fault, without operating any device further upstream.

5.3.2 The discrimination study shall be submitted at S2 detailed design, using IEC 60909 fault current calculations, and shall be re-verified against as-built settings before L4 commissioning of each Section.

5.3.3 No protective device setting shall be finalised without a witnessed relay test recorded in the L3 commissioning documentation.

5.3.4 Arc-flash hazard analysis shall be carried out for all LV and HV switchgear, with hazard labelling fitted before energisation.

5.4 Alarm schedule (extract)

Alarm	Priority	Action
Transformer fault	Critical	Auto-transfer to remaining transformer, DCIM log
Generator fail to start	Critical	Rotate to next available unit, NOC notification
UPS module fault	Critical	Load transfers to remaining N+1 capacity, DCIM log
Battery room off-gas/thermal-runaway detection	Critical	Isolate affected module, initiate fire strategy per PDC-SPEC-07
Rack PDU THD out of band	Warning	Investigate load profile, DCIM trend review

6. Testing and commissioning

Level	Activity	Electrical scope
L1	Factory acceptance testing	Witnessed for HV switchgear, transformers, UPS and generators against schedule ratings
L2	Site delivery and static inspection	Visual inspection, insulation resistance testing, nameplate verification
L3	Individual component start-up and functional test	Each generator, UPS module, switchgear panel started and functionally tested; protection relay testing per Section 5.3
L4	Integrated systems testing, including black-building and failure-mode tests	Loss-of-utility test, loss-of-UPS-module test, loss-of-generator test per PDC-TA-001 §4
L5	Integrated load-bank testing to full Section load for 24 hours continuous	10-second generator load acceptance test at full block load, logged throughout; Section 1 load-banked to full 10 MW shared infrastructure capacity per PDC-BOD-001 §9

6.1 10-second generator load acceptance test

The Subcontractor shall demonstrate, at L4/L5 for each Section, that on simulated loss of the 110 kV incoming supply, the standby generation compound accepts full block load and reaches stable rated voltage and frequency within 10 seconds, with the UPS carrying the load through the transfer with no perceptible transient at the rack PDU. For Section 1, this test shall be performed at the full 10 MW site rating using temporary load banks connected at the hall substations, per PDC-BOD-001 §9. The test shall be repeated with one generator taken out of service to demonstrate N+1 at the generation level, per PDC-TA-001 §4.

7. Submittals

7.1 The Subcontractor shall submit, at the design stages defined in PDC-ER-001 §10:

- single-line diagrams for HV, LV, UPS and generation, at S1 concept and S2 detailed design;
 - short-circuit and protection discrimination study per Section 5.3;
 - arc-flash hazard analysis and labelling schedule;
 - battery fire strategy documentation for lithium-ion installation;
 - equipment schedules with full nameplate data for transformers, generators, UPS and switchgear;
 - commissioning test scripts for L1–L5, including the 10-second load acceptance test script.
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8. Spares, training and warranties

8.1 The Subcontractor shall supply a spares schedule covering not less than two years of routine maintenance consumables (filters, belts, fuel filters) and one critical spare of each protective relay type and UPS power module type used across the site.

8.2 The Subcontractor shall provide not less than 5 days of operator training per Section, covering normal operation, black-building response, generator start sequence and battery room safety procedures, with training records forming part of the S4 handover documentation.

8.3 Equipment warranties shall run for not less than 24 months from the relevant Section's Service Commencement date, or 30 months from delivery, whichever is later, and shall be assignable to the Employer's facilities management provider.

8.4 The Subcontractor shall maintain critical-plant response cover (generators, UPS, HV switchgear) for the defects period, with a maximum 4-hour attendance time for a Critical alarm per Section 5.4.

End of PDC-SPEC-06 Rev A.
